ABHISHEK SINGH KUDAVALE

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Hyderabad (Willing to relocate)
GitHub
LinkedIn
Portfolio

PROFILE

AI/ML Engineer with 2.5+ years of experience specializing in Computer Vision and LLMs. Currently driving AI innovation at IAV GmbH (Germany), deploying scalable AWS microservices and optimizing transformer architectures. Committed to transforming complex algorithms into reliable, high-performance software solutions.

Technical Skills: Python, C++, PyTorch, NumPy, Pandas, OpenCV, Scikit-learn, Huggingface, LangChain, ChromaDB, FastAPI, AWS, Git, Docker

PROFESSIONAL EXPERIENCE

AI Researcher (Master Thesis), *IAV GmbH* — 05/2025 – Present | Gifhorn, Germany

- Optimizing scenario extraction through a multi-layer fusion of computer vision (CV) and vision language models (VLM), currently tracking a 23% increase in accuracy compared to internal baselines.
- Collaborating with cross-functional teams to deploy a low-latency AWS microservice connected to CARLA and dSPACE simulators for automated scenario extraction and generation.
- Developing a SaaS-based benchmarking framework to evaluate CV and VLM models across diverse public and private datasets.
- Designing integrated reasoning systems combining VLM capabilities with CV outputs through prompt optimization and temporal context to enhance semantic scene understanding.
- Implementing and optimizing benchmarked CV models on real-world off-road data, achieving significant improvements in vision tasks.
- Engineering an AI framework integrating computer vision and vision language models for scenario generation for the IAV Mela project.

AI/ML Research Assistant, *Universität Siegen* — 10/2023 – 11/2024 | Siegen, Germany

- Enhanced 6D pose estimation using deep learning and Kalman filters, achieving a 22.88% performance gain in the TRUST-E project.
- Optimized parameters and algorithm configurations to maximize tracking accuracy on real-world IMU data.
- Developed visualization tools to analyze sensor noise profiles and trajectory deviations, enabling data-driven decisions for algorithm tuning.
- Implemented robust preprocessing techniques for IMU signals that significantly reduced drift in challenging conditions.
- Designed reproducible evaluation workflows to quantify improvements from noise tuning and preprocessing.

Graduate Engineering Trainee, *Technique Design Group* — 10/2020 – 11/2021 | Hyderabad, India

- Designed 3D models and assemblies in CATIA V5, ensuring manufacturability and performance optimization.
- Analyzed fabrication metrics and defect rates to identify root causes, driving design improvements that enhanced production yield.
- Deployed ML models (XGBoost, SVMs) on Raspberry Pi, achieving 92% accuracy for real-time 3D CAD analysis.
- Managed fabrication workflows and quality control for client projects from concept to delivery.
- Collaborated with a 5-member cross-functional team to deliver projects on time and within scope.

EDUCATION

Masters of Science – Mechatronics — 10/2022 – Expected 2026 | Siegen, Germany
Universität Siegen; CGPA: 1.9 (Best: 1.0)

Bachelor of Engineering – Mechanical Engineering — 08/2016 – 05/2020 | Hyderabad, India
Vasavi College of Engineering; CGPA: 8.69 (Best: 10.0)

PROJECTS

Advanced RAG for Research Paper Question Answering — 10/2025 – Present
Proficiencies: RAG Pipeline, ChromaDB, LangChain, Huggingface, Prompt Optimization, BERT, Transformers

- Architected an advanced RAG pipeline with reranking, memory integration, and DSPy-based prompt optimization for research paper queries.
- Developed custom question–answer datasets for enhanced context relevance and answer evaluation.
- Implemented evaluation metrics achieving 62% precision using the Gemini-2.5 Flash model.

Fine-Tuning Large Language Model for Spam Classification — 02/2025 – 04/2025
Proficiencies: LLM Pipeline, Token Embedding, Attention Mechanism, Transformer Architecture, LoRA, GPT2

- Fine-tuned a pretrained GPT-2 model (12 attention heads, 117M parameters) using LoRA adapters for spam classification.
- Optimized token embeddings and attention mechanisms through hyperparameter tuning to improve classification performance.
- Achieved 97.21% training and 95.67% test accuracy on 10,000+ spam email dataset.

Robustness and Fairness in Deep Learning — 05/2024 – 07/2024
Proficiencies: CNN, Python, Adversarial Training, Computer Vision, Transfer Learning, Classification

- Performed PGD training with 3 robust loss rules, achieving 84% and 76% accuracy against targeted and untargeted attacks.
- Enhanced security for vision-based systems through adversarial robustness improvements.
- Developed batch sampling techniques for equal class representation, boosting fairness and reducing vulnerability in production AI systems.

ACHIEVEMENTS

- Ranked 1st in 2nd Year of Mechanical Engineering.
- Academic scholarship in Master's program.

LANGUAGES

English, Hindi, Telugu, German